

Name: Alex's Corks

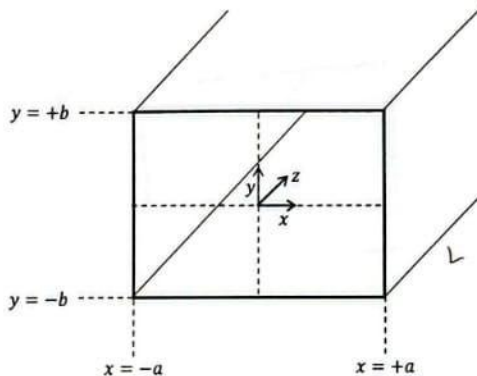
TRANSPORT PHENOMENA II, CHE 311

QUIZ 3

CLOSED BOOK, TIME: 10 MINUTES

A fluid with constant density ρ and viscosity μ flows through a rectangular duct of length L . The modified pressure at the inlet and the outlet of the duct are P_0 and P_L .

- 1- Write the partial differential equation (PDE) that describes the steady-state velocity profile of the fluid.
- 2- Write four boundary conditions needed to solve the above PDE.



90/100

THE EQUATION OF MOTION FOR A NEWTONIAN FLUID WITH CONSTANT ρ AND μ

$$v_z = v_z(x, y)$$

$$[\rho D\mathbf{v}/Dt = -\nabla p + \mu \nabla^2 \mathbf{v} + \rho \mathbf{g}]$$

Cartesian coordinates (x, y, z) :

$$\rho \left(\frac{\partial v_x}{\partial t} + v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y} + v_z \frac{\partial v_x}{\partial z} \right) = -\frac{\partial p}{\partial x} + \mu \left[\frac{\partial^2 v_x}{\partial x^2} + \frac{\partial^2 v_x}{\partial y^2} + \frac{\partial^2 v_x}{\partial z^2} \right] + \rho g_x \quad (\text{B.6-1})$$

$$\rho \left(\frac{\partial v_y}{\partial t} + v_x \frac{\partial v_y}{\partial x} + v_y \frac{\partial v_y}{\partial y} + v_z \frac{\partial v_y}{\partial z} \right) = -\frac{\partial p}{\partial y} + \mu \left[\frac{\partial^2 v_y}{\partial x^2} + \frac{\partial^2 v_y}{\partial y^2} + \frac{\partial^2 v_y}{\partial z^2} \right] + \rho g_y \quad (\text{B.6-2})$$

$$\rho \left(\frac{\partial v_z}{\partial t} + v_x \frac{\partial v_z}{\partial x} + v_y \frac{\partial v_z}{\partial y} + v_z \frac{\partial v_z}{\partial z} \right) = -\frac{\partial p}{\partial z} + \mu \left[\frac{\partial^2 v_z}{\partial x^2} + \frac{\partial^2 v_z}{\partial y^2} + \frac{\partial^2 v_z}{\partial z^2} \right] + \rho g_z \quad (\text{B.6-3})$$

$$v_z(x, y=0) = 0$$

$$v_z(x, y=-b) = 0$$

$$v_z(x=a, y) = 0$$

$$v_z(x=-a, y) = 0$$

$$\boxed{-\frac{\partial^2 v_z}{\partial y^2} = \frac{\partial^2 v_z}{\partial x^2}}$$

$$\frac{\partial p}{\partial z} = 0$$